

Method Applicability Module

Module: MAPM — Method Applicability Module

Parent Standard: PMES — Prediction Methodology Standard

Architecture: PGA™ — Prediction Governance Architecture

Category: Governance & Enforcement

Subcategory: Prediction Governance

Governed Space: Method Applicability

Primary Governed Object: Method Applicability State

Module Type: Core Module

Canonical Language: UCL — Universal Canonical Language

Protection: MIP® — Methodological Intellectual Property

1. Canonical Definition

The Method Applicability Module (MAPM) governs the determination, limitation, conditionality, reassessment, and preservation of whether a defined Prediction Methodology is methodologically applicable to the specific governed predictive frame and conditions in which it is intended to operate.

MAPM determines whether the selected method, defined methodological structure, and documented predictive logic may legitimately be applied to the Prediction Object, Predictive Purpose, Prediction Target, Prediction Horizon, and Predictive Boundary for which the prediction is intended.

A methodology may be technically functional, historically successful, widely accepted, or capable of generating predictive output while remaining methodologically inapplicable to a particular prediction.

Method Capability ≠ Method Applicability

2. Operational Role

MAPM is the fourth Core Module within PMES.

It receives:

Selected Predictive Method

Defined Methodological Structure

Documented Predictive Logic

and evaluates whether that methodology is applicable to the governed predictive frame.

Its principal transition is:

Governed Prediction Methodology Candidate → Applicability Determination → Applicable / Conditionally Applicable / Not Applicable

MAPM does not select the method.



It determines whether the methodology that has been established may legitimately operate within the intended predictive context.

3. Internal Module Flow

Predictive Method Selection → Methodological Structure Definition → Predictive Logic Documentation →
→ Method Applicability → Methodology Version Governance

Current Position: Method Applicability

The governed applicability determination proceeds into:

MVGM — Methodology Version Governance Module

4. Operational Space

MAPM operates within the Method Applicability space.

It governs:

applicability criteria; predictive-frame applicability; purpose applicability; target applicability; horizon applicability; boundary applicability; contextual applicability; methodological preconditions; methodological limitations; conditional applicability; applicability exclusions; applicability dependencies; applicability reassessment; applicability state.

MAPM governs whether the methodology belongs in the predictive context in which it is intended to operate.

5. Operational Scope

Method Applicability may be evaluated for methodologies involving:

statistical forecasting; mathematical modeling; econometric methods; machine learning; AI systems; expert forecasting; human judgment; prediction markets; ensembles; simulation-supported prediction; multi-agent forecasting; hybrid methodologies; other predictive mechanisms.

No methodological class is universally applicable.

6. Applicability Criteria

MAPM requires the criteria governing applicability to be sufficiently identifiable.

Criteria may concern:

Prediction Object characteristics; Predictive Purpose; Prediction Target; Prediction Horizon; Predictive Boundary; methodological requirements; necessary conditions; structural dependencies; information requirements; temporal relevance; environmental conditions; domain limitations; operational constraints.

Applicability must be established against the prediction being governed.

7. Prediction Object Applicability

A methodology suitable for one type of Prediction Object may not be suitable for another.

A method designed for financial time series, for example, does not become applicable to biological outcomes, infrastructure failure, geopolitical events, or human behavior merely because it can mathematically generate an output.

Ability to Process an Object ≠ Applicability to Predict That Object

8. Predictive Purpose Applicability

The same method may be appropriate for one Predictive Purpose and inappropriate for another.

A method suitable for exploratory forecasting may not be appropriate for:

safety-critical decisions; capital allocation; operational control; public policy; high-consequence autonomous action.

MAPM therefore evaluates applicability relative to the purpose established under POPS.

Method Applicability Is Purpose-Dependent

9. Prediction Target Applicability

The methodology must be capable of addressing the specified Prediction Target at the required level of specificity.

A methodology capable of predicting a broad trend may not be applicable to predicting:

a precise value; a specific event; a threshold crossing; a categorical state; a particular operational outcome.

Broad Predictive Capability $\neq$ Target-Specific Applicability

10. Prediction Horizon Applicability

Predictive methods may have different legitimate temporal ranges.

A methodology performing effectively over hours or days may become unreliable over months or years.

Conversely, a long-term trend model may not be suitable for short-term operational prediction.

MAPM therefore requires the methodology to be assessed against the governed Prediction Horizon.

Method Validity at One Horizon $\neq$ Applicability at Another Horizon

11. Predictive Boundary Applicability

A method may be applicable within one geographic, population, organizational, technical, market, environmental, or operational boundary and inappropriate outside it.

Applicability must therefore remain anchored to the Predictive Boundary established by POPS.

Applicability Within Boundary A $\neq$ Applicability Within Boundary B

12. Contextual Applicability

Predictive methodologies operate within conditions.

A methodology developed under one operational, economic, technological, social, environmental, or institutional context may not remain applicable when those conditions materially differ.

MAPM requires contextual applicability to be considered where context materially affects predictive operation.

Methodological Success in One Context $\neq$ Universal Applicability

13. Methodological Preconditions

Some methodologies require specific conditions before they can legitimately operate.

Preconditions may include:

minimum observation history; sufficient event frequency; stable relationships; defined variables; specific measurement conditions; access to required capabilities; functioning dependencies; relevant expertise; other methodological prerequisites.

Where a required precondition is absent, applicability may become conditional or fail.

14. Information Availability and Applicability

A methodology may require certain categories, quantities, continuity, or timeliness of information.

MAPM may identify such information requirements as methodological applicability conditions.

It does not determine whether the actual information used forms a sufficient governed Prediction Basis.

That belongs to PBAS.

Method Information Requirement $\neq$ Prediction Basis Sufficiency

15. Assumption Dependency and Applicability

A methodology may depend upon assumptions about the environment in which it operates.

MAPM may establish that applicability depends upon such conditions.

PAS governs the assumptions themselves.

Method Applicability Condition $\neq$ Governed Predictive Assumption

A methodology must not be represented as unconditionally applicable where its operation materially depends upon conditional premises.

16. Methodological Limitations

Known limitations capable of restricting applicability must remain identifiable.

Limitations may concern:

domain; scale; horizon; population; data conditions; volatility; structural change; rare events; extreme outcomes; environmental stability; model behavior; human judgment; computational constraints.

A limitation does not automatically invalidate a method.

It establishes a boundary within which applicability must be interpreted.

Methodological Limitation $\neq$ Methodological Failure

17. Applicability Exclusions

A methodology may have circumstances in which it must not be treated as applicable.

Where such exclusions are materially known, they should be explicitly governable.

Examples may include:

unsupported populations; unsupported horizons; environments outside training or historical conditions; structurally different markets; unavailable required variables; extreme conditions outside methodological capability.

Known Exclusion $\neq$ Acceptable Applicability by Silence

18. Conditional Applicability

A methodology need not always be classified simply as applicable or not applicable.

It may be:

Applicable only if specified conditions hold.

Conditional applicability must identify the conditions upon which the determination depends.

This allows governance to distinguish:

Applicable

from:

Applicable Only Under Defined Conditions

19. Partial Applicability

A methodology may be applicable to only part of the governed predictive problem.

For example, it may legitimately predict a broad directional trend but not the precise magnitude required by the Prediction Target.

Or it may apply to part of a population but not the entire Predictive Boundary.

Partial applicability must not be represented as full applicability.

Partial Applicability $\neq$ Full Applicability

20. Applicability and Method Complexity

Complexity does not establish applicability.

A sophisticated AI system, large ensemble, advanced simulation, or computationally intensive model may remain inappropriate for the predictive problem.

Likewise, a comparatively simple method may be applicable where the predictive problem does not require greater complexity.

Methodological Complexity $\neq$ Method Applicability

21. Applicability and Historical Performance

Historical success may provide relevant information but cannot independently establish present applicability.

Performance may have occurred:

under different conditions; across another horizon; for another target; within another population; under another methodology version; before structural changes in the predicted environment.

Historical Accuracy $\neq$ Current Method Applicability

Predictive performance governance belongs to POFPS.

22. Applicability and Benchmark Performance

Benchmark performance may demonstrate capability under benchmark conditions.

It does not automatically establish applicability to a real predictive context.

This distinction is particularly important for AI and machine-learning systems.

Benchmark Performance $\neq$ Operational Predictive Applicability

A model may perform strongly on standardized evaluations while lacking sufficient applicability to the specific prediction being governed.

23. Applicability and Predictor–Reality Proximity

A predictor or predictive source may possess unusually strong access to information, operational actors, decision processes, or conditions from which the outcome emerges.

Such proximity may improve observed predictive performance without establishing that the underlying methodology is generally applicable.

MAPM must therefore avoid treating information advantage or proximity to Operational Reality as automatic evidence of methodological applicability.

Information Advantage $\neq$ Method Applicability

Outcome Knowledge $\neq$ Predictive Capability

Predictor–Reality Proximity is governed elsewhere in PGA™ where its source and performance implications become relevant.

24. Prediction Market Applicability

A prediction market may be applicable where market-based aggregation legitimately serves the Predictive Purpose.

However, market liquidity, participant count, trading activity, or availability of a market price do not independently establish methodological applicability.

The market may contain:

concentrated influence; dependent participants; information asymmetry; outcome-informed participants; limited representation; strategic trading.

These characteristics require governance elsewhere in PGA™.

MAPM governs only whether use of the market mechanism is methodologically applicable to the predictive context.

Active Market ≠ Applicable Prediction Method

25. AI Method Applicability

An AI model may have broad general capabilities while lacking prediction-specific applicability.

MAPM requires applicability to be assessed against the governed predictive frame rather than inferred from model reputation, scale, benchmark performance, or general intelligence.

General AI Capability ≠ Prediction-Specific Applicability

This distinction becomes increasingly important as general-purpose models are used for specialized forecasting tasks for which they were not specifically designed or evaluated.

26. Applicability Under Changed Conditions

Method Applicability may change.

Material changes may include:

Prediction Object change; Purpose change; Target change; Horizon change; Boundary change; methodological dependency change; environmental regime change; technological change; structural market change; loss of required information; emergence of previously unknown limitations.

A prior applicability determination must not automatically survive materially changed conditions.

Historical Applicability ≠ Continuing Applicability

27. Applicability Reassessment

Where a material applicability condition changes, MAPM requires reassessment.

The result may be:

applicability preserved; applicability restricted; applicability made conditional;

applicability suspended; methodology found not applicable.

Reassessment must not silently rewrite the historical applicability determination.

28. Applicability Sufficiency

Method Applicability is sufficiently governed when:

applicability criteria are identifiable; the methodology is assessed against the complete governed predictive frame; material preconditions are identified; relevant limitations are recognized; exclusions are identified where known; conditional or partial applicability is represented where appropriate; dependencies are identifiable; changed conditions can trigger reassessment; the applicability state is explicitly established.

Applicability does not establish predictive correctness.

29. Governance Requirements

MAPM requires that:

1. Method Applicability be assessed against the governed predictive frame; 2. applicability criteria remain identifiable; 3. Prediction Object applicability be established; 4. Purpose applicability be considered; 5. Target applicability be established; 6. Horizon applicability be established; 7. Predictive Boundary applicability be established; 8. material methodological preconditions remain identifiable; 9. material limitations and exclusions be recognized; 10. conditional or partial applicability not be represented as unconditional applicability; 11. historical or benchmark performance not substitute for applicability determination; 12. material changes trigger applicability reassessment; 13. the resulting applicability state remain traceable.

30. Governance Logic

MAPM applies the following operational logic:

Selected Predictive Method + Defined Methodological Structure + Documented Predictive Logic

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Applicability Criteria Established

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Prediction Object Applicability

↓

Purpose Applicability

↓

Target Applicability

↓

Horizon Applicability

↓

Predictive Boundary Applicability

↓

Preconditions / Dependencies Assessed

↓

Limitations / Exclusions Assessed

↓

Conditional / Partial Applicability Determined where relevant

↓

Applicability State Determined

↓

MVGM — Methodology Version Governance Module

Where sufficient applicability cannot be established:

Applicability Governance Gap → Clarification / Restriction / Alternative Method / Reassessment

31. Minimum Implementation Framework

A minimum implementation of MAPM must establish:

A. Methodology Reference

The selected method, defined structure, and documented predictive logic being assessed.

B. Predictive Frame Reference

The applicable POPS-governed predictive frame.

C. Applicability Criteria

The criteria against which applicability is determined.

D. Frame Applicability

Applicability to:

Prediction Object · Predictive Purpose · Prediction Target · Prediction Horizon · Predictive Boundary

E. Preconditions

Material conditions required for legitimate methodological use.

F. Dependencies

Dependencies capable of materially affecting applicability.

G. Limitations

Known methodological limitations relevant to the predictive frame.

H. Exclusions

Known conditions under which the methodology must not be treated as applicable.

I. Conditionality

Conditions restricting or qualifying applicability.

J. Reassessment Triggers

Material changes requiring applicability reconsideration.

K. Applicability State

At minimum:

Undetermined → Under Assessment → Applicable → Conditionally Applicable → Partially Applicable → Reassessment Required → Not Applicable

Only a methodology with a sufficiently governed applicability state may proceed as an applicable Prediction Methodology.

32. Governance Outputs

MAPM may produce:

Method Applicability Determination; applicability criteria; predictive-frame applicability profile; methodological preconditions; dependency conditions; limitation record; exclusion conditions; conditional

applicability requirements; partial applicability boundary; reassessment triggers; applicability state; unresolved applicability requirements.

The primary governed output is:

Method Applicability State

This becomes a principal input into:

MVGM — Methodology Version Governance Module

33. Critical Distinctions

Method Capability $\neq$ Method Applicability

Method Availability $\neq$ Method Applicability

Ability to Process an Object $\neq$ Applicability to Predict That Object

Method Applicability Is Purpose-Dependent

Broad Predictive Capability $\neq$ Target-Specific Applicability

Method Validity at One Horizon $\neq$ Applicability at Another Horizon

Applicability Within Boundary A $\neq$ Applicability Within Boundary B

Methodological Success in One Context $\neq$ Universal Applicability

Method Information Requirement $\neq$ Prediction Basis Sufficiency

Method Applicability Condition $\neq$ Governed Predictive Assumption

Methodological Limitation $\neq$ Methodological Failure

Partial Applicability $\neq$ Full Applicability

Methodological Complexity $\neq$ Method Applicability

Historical Accuracy $\neq$ Current Method Applicability

Benchmark Performance $\neq$ Operational Predictive Applicability

Information Advantage $\neq$ Method Applicability

Outcome Knowledge $\neq$ Predictive Capability

Active Market $\neq$ Applicable Prediction Method

General AI Capability $\neq$ Prediction-Specific Applicability

Historical Applicability $\neq$ Continuing Applicability

34. Enterprise Application

MAPM prevents organizations from confusing access to a predictive technology with legitimate applicability of that technology to a predictive problem.

This is increasingly important as general-purpose AI systems, forecasting platforms, prediction markets, simulation systems, and commercially available predictive tools can generate outputs across domains far beyond the environments in which their predictive capability has actually been established.

MAPM creates the governance checkpoint between:

"This method can produce a prediction."

and:

"This method is applicable to this prediction."

35. Use Case 1 — High-Accuracy Model Deployed Into a Different Reality

A predictive model demonstrates excellent historical performance in one market, country, population, or operational environment.

The organization then deploys the same methodology into another environment because the Prediction Target appears similar.

The model continues to generate precise forecasts.

However, the new environment contains different behavioral patterns, institutional conditions, operational constraints, market structures, or population characteristics.

No formal applicability reassessment occurs.

The governance question is:

Can historical predictive success in one environment justify using the same methodology in another environment when the conditions that made the method effective may no longer exist?

MAPM requires applicability to be established against the actual predictive frame and operating conditions.

Historical Success $\neq$ Transferable Applicability

36. Use Case 2 — General-Purpose AI Used for a Safety-Critical Prediction

A general-purpose AI system is asked to predict whether a critical infrastructure component will fail within the next 30 days.

The system can analyze technical documentation, maintenance records, historical failures, and operational descriptions and produces a confident numerical probability.

The model is highly capable and performs strongly on broad benchmarks.

But the organization cannot establish whether its methodology is applicable to this specific equipment class, failure mechanism, operating environment, Prediction Horizon, or safety-critical Predictive Purpose.

The governance question is:

Should a prediction be relied upon merely because a powerful AI system can generate it when prediction-specific methodological applicability has never been established?

MAPM separates general capability from prediction-specific applicability.

The ability to answer the predictive question does not establish the methodological right to answer it.

37. Canonical Boundary

MAPM governs the determination, limitation, conditionality, partial applicability, exclusions, dependencies, reassessment, and state of Method Applicability relative to a governed predictive frame. It does not govern initial method selection, methodological structure definition, predictive logic documentation, methodology version continuity, Prediction Basis sufficiency, assumptions, prediction formation, probability representation, feasibility, collective prediction legitimacy, Operational Reality, or predictive performance.

MAPM establishes whether the defined Prediction Methodology is legitimately applicable to the prediction for which it is intended .

It does not establish whether that prediction will ultimately be correct .

Canonical Closing Statement

The ability of a methodology to produce a prediction does not establish its applicability to the prediction being governed.

MAPM requires predictive capability to become contextually applicable before it can become methodologically legitimate.

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